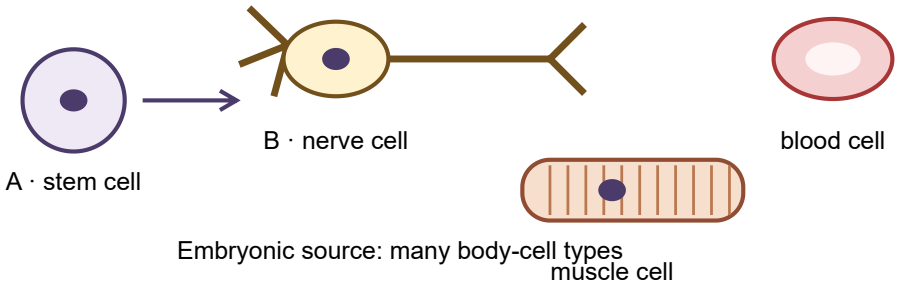
Reference A animal cells



An embryonic stem cell can give rise to many specialised body-cell types.

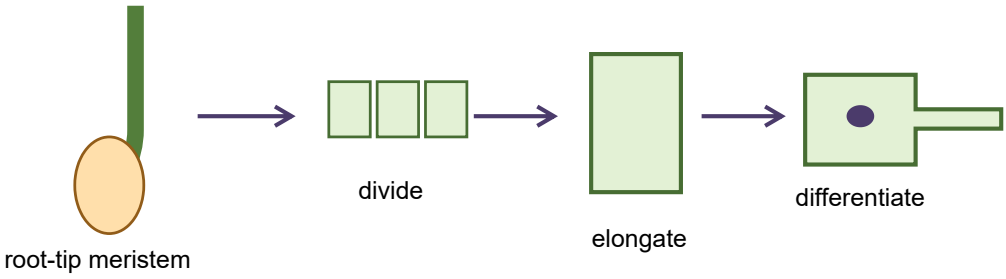
Reference only. Do not write on this sheet. Trace a feature, then explain its function.

Cell or source	Feature or range	Function or limit
Nerve cell	Long extension	Carries signals over a distance
Muscle cell	Structures that contract	Contraction produces force
Embryonic stem cell	Can give rise to many body-cell types	Not a promise of a successful treatment
Adult blood-forming stem cell	More limited range of blood-cell types	Makes replacement blood cells

The arrows show possible lineages. The drawing does not show every cell stage. Cells do not choose their own jobs. Specialised cells can share inherited DNA but use it differently.

Made by Matt

Reference B root growth



Root-tip meristem, division, elongation and differentiation in a simplified route.

Reference only. Do not write on this sheet.

Process or region	What it adds	Useful limit
Meristem	A region where cells divide	Not every root cell remains a meristem cell
Division	More cells	Number alone does not prove a specialised role
Elongation	Longer cells	Not the name for becoming specialised
Differentiation	Features suited to a job	Different cells can be at different stages

A root hair extension provides a large surface for absorption. The route is simplified. Use it to compare processes, not as a claim that every cell follows an identical sequence.

Made by Matt

Arrival and exit Aspire

Use the labelled picture on Reference A for question 1. Respond in your usual mode.

1. Use the picture only. Give the letter and label of the cell that can give rise to specialised types.

2. State what is copied before mitosis and what cell splitting produces.

3. A tissue has more cells but no new cell types. Which process is shown? Which extra process would explain specialised roles?

4. Why can a shoot become longer after cells have been made in a meristem?

Exit ticket

1. Why does more cells not by itself show a new specialised role?

2. A root lengthens. Give one possible process and one limit of that evidence.

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

Made by Matt

You do Aspire

Use your book or this sheet. One response per task. Use Reference A and B.

1. A model shows two new cells, both labelled specialised. Explain why cell number alone cannot justify those labels. Add the missing process. [3] Staff I ☐ S ☐ C ☐

---

2. Compare a nerve cell and muscle cell. Link one feature of each to its job, then state what a drawing cannot establish. [4] Staff I ☐ S ☐ C ☐

---

3. Evaluate: Every adult stem cell can become any body cell. Correct the scope and use the blood-forming example. [4] Staff I ☐ S ☐ C ☐

---

4. Compare the skin and root routes. Explain two shared ideas and one plant difference. Give one limit of a simple three-step sequence. [5] Staff I ☐ S ☐ C ☐

---

5. Repair: A longer root proves every new cell is a stem cell. State what growth could show, what it does not show and a relevant extra check. [4] Staff I ☐ S ☐ C ☐

Feedback and response · Aspire

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt